Research in soft matter engineering has developed innovative approaches for robotics and wearable devices that can create interfaces compatible with the human body and adapt to unpredictable environmental conditions. However, many promising works in this field remain limited by the dependence of soft systems on electrical or pneumatic tethers. In recent years, studies on soft actuators and flexible electronics have made the removal of these tethers more feasible (Shepherd et al., 2011, as cited in Rich et al., 2018), thus paving the way for a wide range of application sectors, from autonomous field robotics to wireless biomedical devices (Fine et al., 2014, as cited in Rich et al., 2018). In this context, biohybrid approaches, in which biological tissues and cells are integrated with robotic systems, have a significant intersection with the fields of molecular biology and genetics.

Concepts in Soft Robotics

Flexible Materials (Elastomers, Liquid Metals)

The foundation of soft robotic systems is built on special materials like elastomers and liquid metals, which exhibit high flexibility and deformation capabilities. Elastomers can stretch to great lengths and return to their original form due to their polymer structures, allowing them to safely and effectively mimic muscle-like movements (Shepherd et al., 2011, as cited in Rich et al., 2018). Silicone-based elastomers (e.g., PDMS, Ecoflex) are frequently preferred in both wearable systems and biohybrid applications that come into contact with cells, thanks to their biocompatibility. On the other hand, liquid metals—especially EGaIn (eutectic gallium-indium) (Amjadi et al., 2016, as cited in Rich et al., 2018)—are used to produce stretchable circuits, sensors, and actuator systems by being integrated into elastomers as microchannels. This is due to their high electrical conductivity and their liquid state at room temperature. These materials enable both the structural flexibility and electronic functionality of soft robots, offering an ideal infrastructure for applications particularly in the biomedical field.

Actuation Methods (Pneumatic, Thermoelectric, Light-sensitive Polymers)

Actuation is one of the main challenges in achieving autonomous soft robots, and this issue is addressed with various actuation methods such as pneumatic, thermal, light-sensitive, biohybrid, electrostatic, and electro-osmotic (Shepherd et al., 2011, as cited in Rich et al., 2018). Each method has its own unique advantages and limitations in terms of actuation force, speed, environmental compatibility, and energy requirements. Especially in biomedical and wearable applications, it is important to prefer methods that offer low power consumption and high compatibility (Shepherd et al., 2016, as cited in Rich et al., 2018).

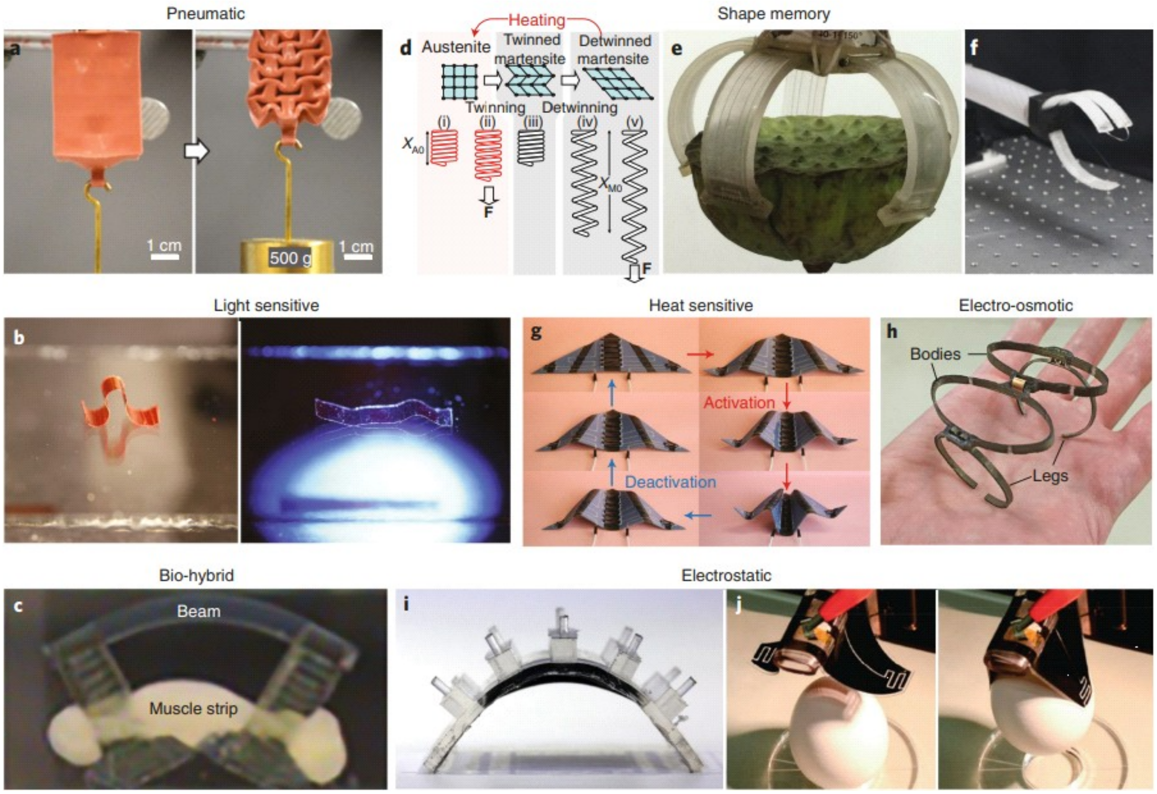

Figure.1 | Methods of soft actuation Different actuation methods used in soft robotic systems: (a) Pneumatic contraction, (b) Light-sensitive movement, (c) Biohybrid movement with muscle cells, (d) Shape-memory alloys, (e) Gripper with thermal heating, (f) Electrostatic gripper, (g) Thermoactive bending, (h) Electro-osmotic movement, (i–j) Activating grippers with an electrostatic field.

Sensor Integration

For soft robots to interact with their environment, they require flexible, stretchable, and biocompatible sensors.

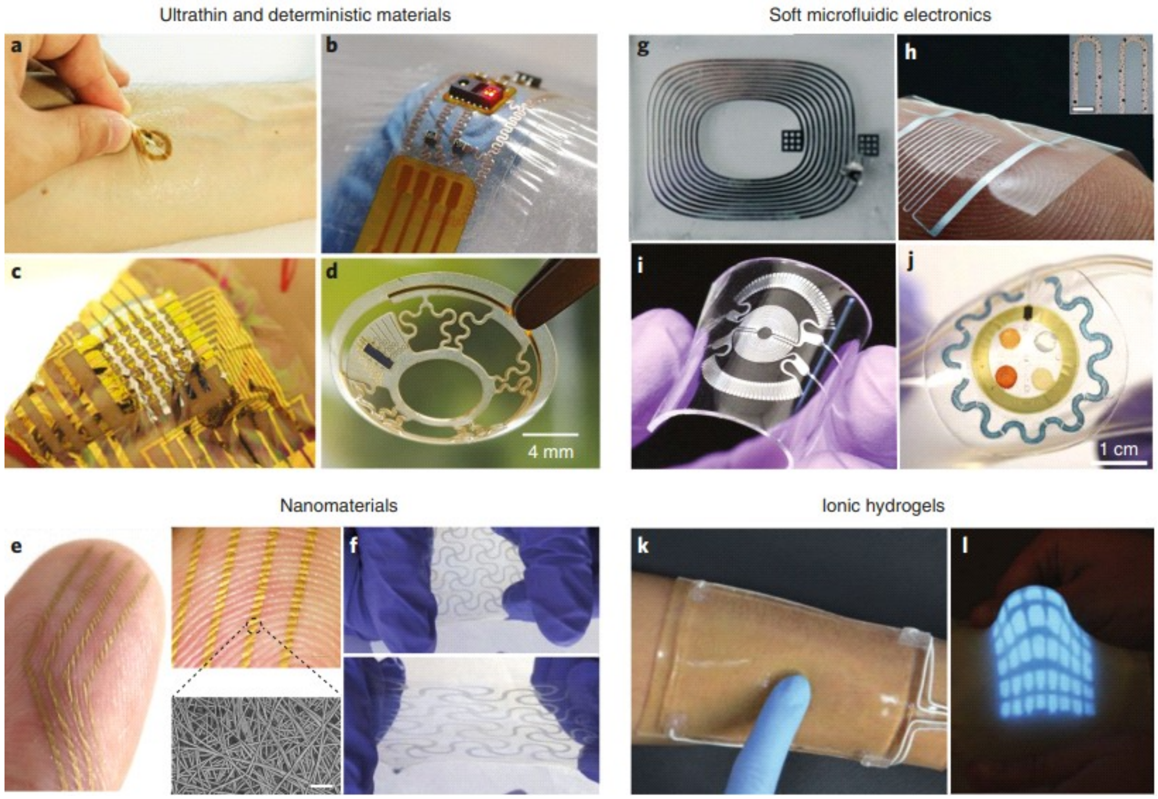

Figure.2 | Advances in Soft Sensing, Conductivity and Artificial Skin. This figure presents various examples of soft sensing and conductive technologies: (a–d) Sensors using ultrathin architectures (e.g., on-skin sensors, integrated contact lenses); (e,f) Conductive structures achieved by nanomaterials (e.g., traces for strain, stretchable PEDOT:PSS film); (g–j) Conductive and wearable devices enabled by microfluidics (e.g., EGaIn antennas, pressure sensors, sweat analysis patches); and (k,l) Wearable devices using ionic hydrogel electronics (e.g., soft touch panels, stretchable LED arrays).

Fully Autonomous Soft Robot Systems

Developing fully autonomous systems in the field of soft robotics requires robot designs that not only have flexible actuators but also integrated power sources, control circuits, and communication systems. The examples in Figure 3 represent current approaches to autonomous soft robots that can operate in various environments and move without the need for external wiring (Shepherd et al., 2011, as cited in Rich et al., 2018).

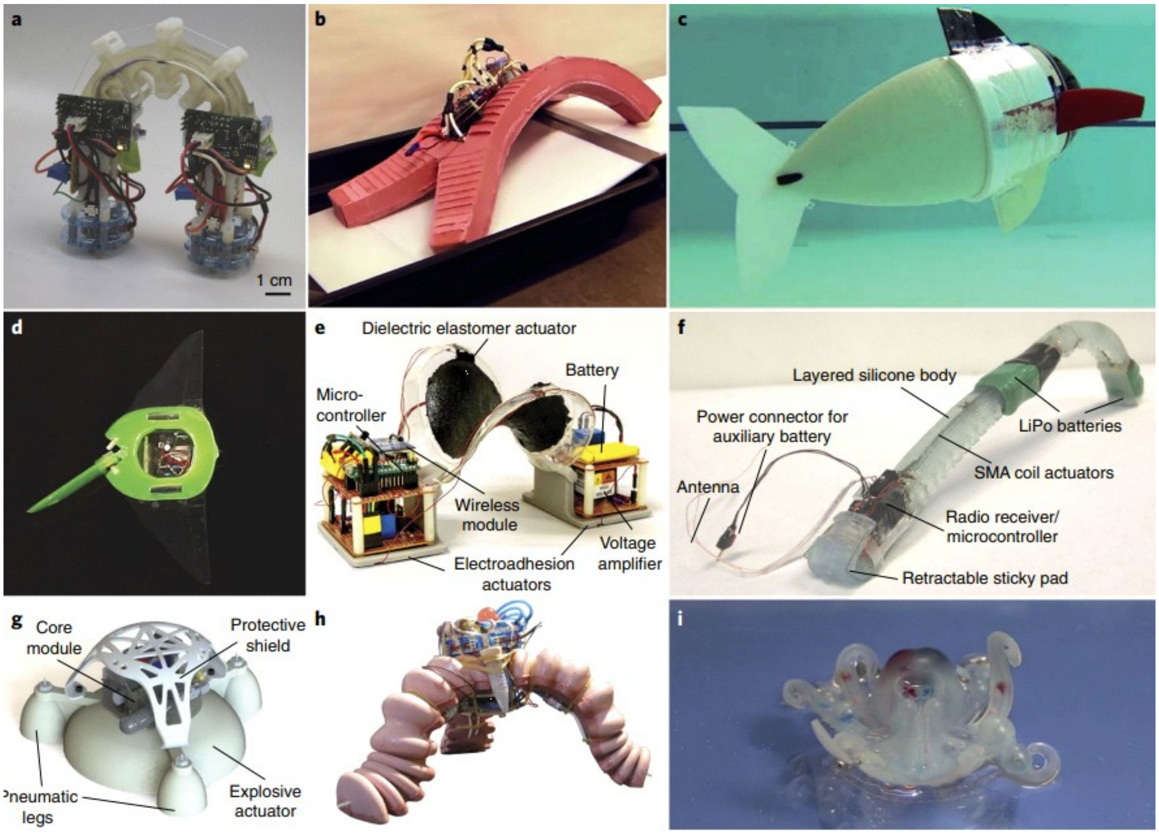

Figure.3| Fully Untethered Robotic Systems. Examples of autonomous soft and hybrid robots with on-board power and control: (a) Motor-cable powered climbing robot; (b) Robust walking robot using on-board pneumatics; (c,d) Biomimetic swimming robots (hydraulic or DEA-powered); (e) DEA-powered walking robot with electro-adhesion; (f) SMA-powered multi-gait caterpillar-inspired robot; and (g–i) Jumping and crawling robots powered by combustion systems, including a fluidic logic-controlled octopus robot.

The robots in images (a) and (b) operate with shape-memory alloys (SMA) and can perform directional crawling motion. The fish robot in image (c) is capable of swimming in a fluid environment and is supported by an integrated hydraulic system. Image (d) shows a phototactic robot that exhibits light-seeking behavior, equipped with dielectric elastomer actuators. (e) is a multifunctional terrestrial robot capable of surface adhesion based on electro-adhesion. Robot number (f) is designed to include SMA coils, adhesive pads, a microcontroller, and a battery within its silicone body, and it has both crawling and gripping functions. In examples (g) and (h), robots that perform sudden jumps and leaps with explosive actuators are seen; these systems are designed for tasks requiring rapid displacement. Finally, (i) shows an electronics-free octopus robot composed of entirely soft materials and powered by a microfluidic-controlled combustion system (Shepherd et al., 2014, as cited in Rich et al., 2018).

Conclusion

Soft robotics is a rapidly developing research area, with recent advances in material science, actuator technologies, and system integration. Innovations in sensing and actuation mechanisms, the use of next-generation materials, and developments in manufacturing techniques are paving the way for more durable, precise, and energy-efficient systems. These advances are not limited to engineering-based solutions; the integration of knowledge about the functioning of biological systems into robotic systems is also becoming increasingly important.

In particular, biohybrid approaches, by integrating cell-based actuators and biological sensors into robotic platforms, are enhancing the functionality of soft robots and creating new application areas. In this context, topics such as cell culture, tissue-engineered structures, and genetic-level control mechanisms present a multi-disciplinary field of study that requires a biological knowledge base. Research in molecular biology and genetics provides the necessary biological foundation for modeling and directing living systems, thereby contributing to robotic systems becoming more compatible, sensitive, and customizable with their environment. In the coming period, the joint progress of both engineering and biology-based approaches will enable soft robots to be used more effectively in fields such as healthcare, wearable technology, and environmental interaction.

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